Mahmoud Alyosify

Reinforcement Learning Engineer

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SUMMARY

Reinforcement learning engineer who builds agents for **decisions with a real cost**. Built a cloud-autoscaling environment from scratch and trained PPO and DQN agents to balance cost and latency, compared with the policies teams actually run. Master's thesis treated text generation as a stop-or-continue decision and cut 47.6% of generated tokens with no answer cut off. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), with graduate coursework in reinforcement learning.

SKILLS

Reinforcement learning: PPO, DQN, custom Gymnasium environments, reward design, simulation, policy evaluation
Machine learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
Agents: LangGraph, LangChain, multi-agent systems, tool use
Evaluation: baselines and ablations, statistical testing, reproducible experiments
Engineering: Python, C++, Docker, AWS, Git, Linux

PROJECTS

RL-Cloud-Autoscaler — deep RL for cloud scaling — 2026

Python, Gymnasium, PPO, DQN

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.
- Designed the environment from scratch: state, actions and a reward that penalises both overspending and slow responses.
- Compared both agents with static allocation and threshold-based autoscaling, the policies real teams run.

MinimalLM — Master's thesis, published on arXiv — 2026

Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- Treats generation as a stop-or-continue decision: a stop rule with a completeness gate makes three open LLMs generate **47.6% fewer tokens**, with no answer cut off.
- Found that the simplest stop rule saves 32.6% but cuts off 59.3% of answers; added a guardrail instruction and a completeness check so the saving no longer costs the answer.

HORUS Sentinel — multi-agent threat-intelligence platform — 2026

Python, LangGraph, FastAPI, Neo4j, Ollama, Docker

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.
- Each agent owns one source; every finding is linked to earlier evidence in the shared graph, so reports trace each claim back to its source.

SCRAP — satellite collision risk prediction — 2026

Python, XGBoost, LightGBM, Optuna, SHAP

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.

Also: SimCLR-Vision-SSL (84.30% top-1, self-supervised) · Horus-OSINT (Llama-3-8B fine-tuned on AWS) · Automated Multimodal Agent (PDF to narrated slides).

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Built a **LangGraph** agent that routes each question, grades the evidence it finds, rewrites weak queries and checks its own answer.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

Part-Time Machine Learning Instructor Jul 2025 – Present
National Telecommunication Institute (NTI) Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.

AI in Cybersecurity Intern Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.

Online Instructor 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi); published on arXiv
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- Fundamentals of Reinforcement Learning (University of Alberta, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- IBM AI Developer Professional Certificate (IBM, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)